

UGTS LIBRARY EDITION

# VOLUME 0

## THE MAP OF THE WHOLE JOURNEY

A chronological, layered guide for ages 5-10 and up  
Foundation charter, family tree and first library lesson

1

**WHERE?**

support

2

**MATCH?**

compatibility

3

**READY?**

guard

4

**CHECKED?**

verified event

5

**CHANGE**

transition

6

**REMEMBER**

lineage

# THIS BOOK IS THE MAP, NOT THE WHOLE LIBRARY

Installment 1: chronology, learning routes and a first activity

## Purpose

UGTS has grown through several technical reports, formal profiles and applications. This first library volume untangles that history before any single topic is taught in depth. It is written for children, families, educators and adult learners who benefit from concrete language, visible order and honest boundaries.

### THE LIBRARY PROMISE

Every short child question stays connected to the real formal term. Every release keeps its date, type, parent and limits. Missing documents stay visible. Tested examples are not presented as universal proof.

## What we will not do

- We will not pretend every larger version number belongs to one straight chain.
- We will not turn one-bit flags, drawings or topology words into magical complete explanations.
- We will not hide the missing UGTS 3.6.1 artifact.
- We will not place private identifiers on child-facing pages.
- We will not call a demonstration a finished product or a test suite independent scientific proof.

### ONE-SENTENCE DEFINITION

UGTS is a way to describe what exists, what may interact, when a change is allowed, what change happens, and what must be remembered afterward.

# HOW TO READ THE LIBRARY

A young reader may use formal words; an adult may prefer concrete play

## FOUR READING LANES - NO FIXED AGE GATE

Choose the route that gives the reader access today. Move between lanes freely.

| LOOK & SAY                                                                                  | PLAY & BUILD                                                                           | READ & REASON                                                                                  | STUDY & VERIFY                                                                                                 |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Usually 5+<br><br>Big pictures, short questions, pointing, signing, AAC and shared reading. | Usually 7+<br><br>Cards, objects, sorting, routes, simple rules and cooperative games. | Usually 9+<br><br>Formal words, cause and effect, order, evidence and small design challenges. | Older learner / adult<br><br>Version lineage, definitions, tests, boundaries, source notes and implementation. |

### ACCESS RULE

Age labels are invitations, not gates. A reader may answer by speaking, pointing, signing, eye gaze, switches, objects, drawing or writing. Understanding matters more than the response method.

The same page can serve different readers

| CHILD OR CONCRETE ROUTE                                                                                                                     | ADULT OR FORMAL ROUTE                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Read the large question, handle an object, make a choice and see the result. Formal vocabulary may be introduced naturally, without a test. | Read the source note, distinguish inherited architecture from engineering additions, inspect validation claims and preserve non-claims. |

# WHAT DOES “SUBSTRATE” MEAN?

The rule-ground underneath a system

Imagine a game, a robot, a library checkout desk or a building set. Pictures and sounds show what people see. The substrate is the organized rule-ground that says which objects exist, which rules apply, which interactions are allowed, when an important boundary is crossed, and what history must be kept.

## CHILD VERSION

The substrate is the part underneath that helps the world make fair, repeatable changes.

Three separations protect the idea

### RULES ARE NOT PICTURES

A render, mesh, SVG or screen image may help us see. It does not automatically become the authority for state and change.

### PLACE IS NOT IDENTITY

Two things can share a coordinate but belong to different sheets, phases, routes or histories. Lineage keeps the difference.

### PROPOSAL IS NOT CHANGE

A hand, GPU, sensor or game system may suggest a change. The change becomes authoritative only after the declared checks and deterministic commit.

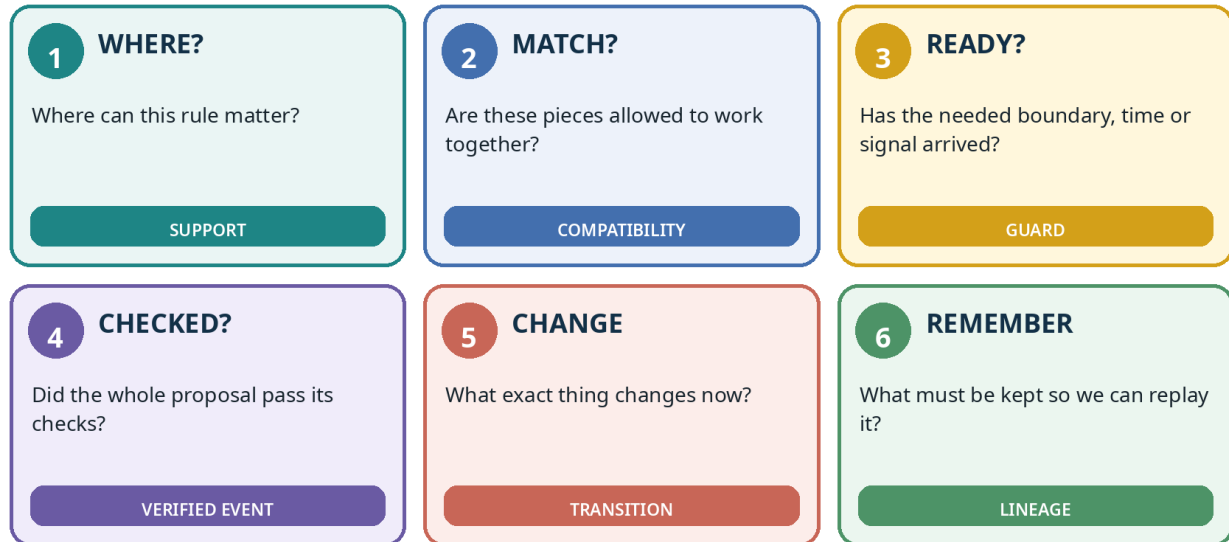
**Source note:** This distinction is repeated across UGTS-GN 1.1, UGTS-KC 2.0, Two Hands 3.0, UGTS-KC 3.6 and KC Elizabeth 3.9.

# THE SIX BIG QUESTIONS

The permanent backbone of the library

## THE SIX BIG QUESTIONS

The short question comes first. The formal word is always nearby.



### SAY THEM IN ORDER

WHERE? -> MATCH? -> READY? -> CHECKED? -> CHANGE -> REMEMBER

**Source note:** Formal basis: the canonical shorthand in UGTS-GN 1.1 is local support -> compatibility -> guard crossing -> verified event -> route/transition -> lineage + novelty log. Later releases retain this authority chain.

# FIRST STORY: THE LIBRARY DOOR

A familiar example of the six questions

Mika wants to borrow a red activity ball from a locked library cupboard. The librarian and Mika use six questions.

|          |                                                                                             |
|----------|---------------------------------------------------------------------------------------------|
| WHERE?   | Is Mika at the right cupboard, and is the red ball inside it?                               |
| MATCH?   | May this library card borrow this kind of ball?                                             |
| READY?   | Is the cupboard available, and has checkout begun?                                          |
| CHECKED? | Did the scanner confirm the card, the ball and the rule?                                    |
| CHANGE   | Open the door and mark the ball as borrowed.                                                |
| REMEMBER | Keep the borrower, item, time, due date and reason so the loan can be returned or replayed. |

## ADULT NOTE

This is an analogy, not a claim that every library uses UGTS. It demonstrates the order: local relevance, permission, threshold, verification, change and record.

# WHY ORDER MATTERS

The same pieces in the wrong order can make a wrong world

## SAFE ORDER

1. Find the relevant place.
2. Check whether the parts may interact.
3. Check the boundary or timing.
4. Verify the whole proposal.
5. Apply one clear change.
6. Record what happened.

## SCRAMBLED ORDER

Change first, then check later.  
Remember an event that did not happen.  
Use a picture as state authority.  
Treat same position as same identity.  
Let two systems write conflicting changes at once.

A small construction challenge

1. Place six paper cards in a line.
2. Ask a partner to swap two cards.
3. Tell a short story that goes wrong because of the swap.
4. Restore the order and add a memory bead for the corrected event.

## FORMAL BRIDGE

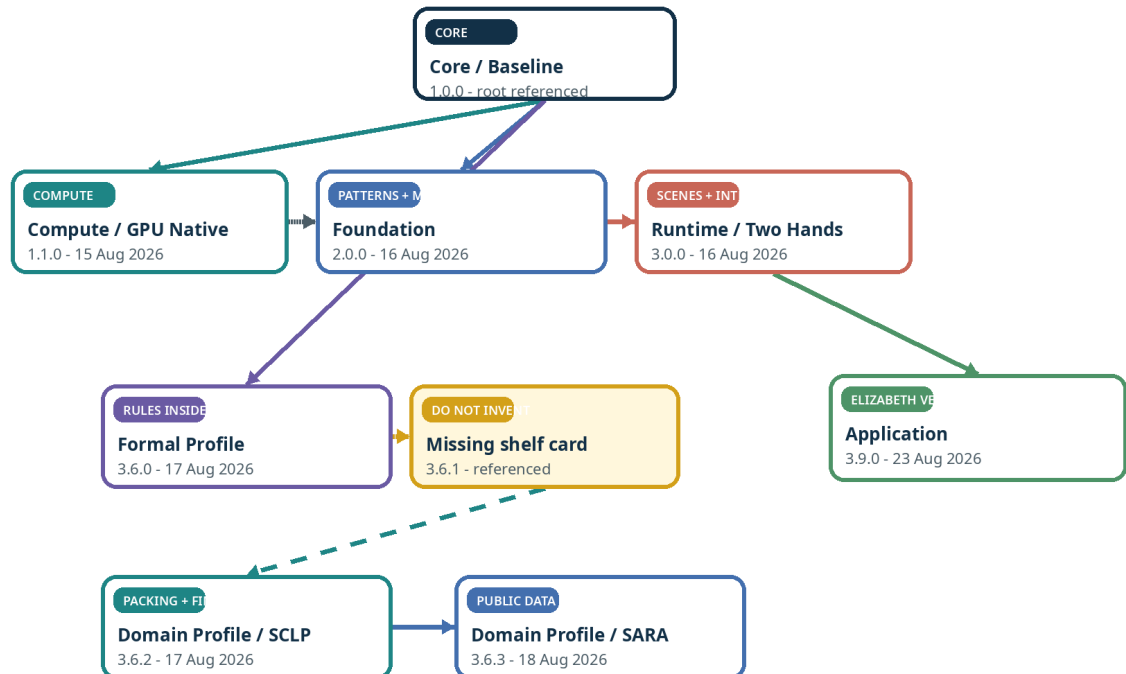
UGTS-KC 3.6 makes operation order explicit inside the substrate: parse, normalize, resolve references, construct, transform, support, compatibility, guard/root, transition and lineage/log.

# THE VERSION NUMBERS ARE A FAMILY TREE

Chronology and lineage are different questions

## UGTS IS A FAMILY TREE, NOT ONE STRAIGHT LADDER

Dates tell when a document appeared. Arrows tell what it actually builds on.



Solid arrows: package lineage. Dashed arrows: referenced, shared or incomplete relationship.

### IMPORTANT

A later date does not mean a document replaces every earlier document. A domain profile may extend one branch; an application may extend another. The library will always print both the release date and the actual parent relationship.

# CHRONOLOGICAL RELEASE LEDGER

The current audited shelf

| DATE               | CANONICAL NAME                                | VERSION | RELATIONSHIP                                     |
|--------------------|-----------------------------------------------|---------|--------------------------------------------------|
| Before 15 Aug 2026 | Core / Baseline Architecture                  | 1.0.0   | Root referenced; source not in current shelf     |
| 15 Aug 2026        | Compute / GPU Native                          | 1.1.0   | Extends and normalizes the root                  |
| 16 Aug 2026        | Foundation / Pattern, Kinematics and Dynamics | 2.0.0   | Builds on core and compute ideas                 |
| 16 Aug 2026        | Runtime / Two Hands                           | 3.0.0   | Builds on the 2.0 foundation package             |
| 17 Aug 2026        | Formal Profile / Literal Referential          | 3.6.0   | Formal branch: definitions become substrate data |
| 17 Aug 2026        | Unresolved 3.6.1 artifact                     | 3.6.1   | Referenced but not supplied                      |
| 17 Aug 2026        | Domain Profile / SCLP                         | 3.6.2   | Builds on the formal branch                      |
| 18 Aug 2026        | Domain Profile / SARA                         | 3.6.3   | Builds on SCLP package discipline                |
| 23 Aug 2026        | Application / Elizabeth Vector Game           | 3.9.0   | Builds on the Two Hands runtime branch           |

## CATALOG HONESTY

The missing 3.6.1 row is deliberately visible. Hiding it would make the chart prettier but the history less truthful.

# 1.1 COMPUTE / GPU NATIVE

The rules learn how to check many candidates carefully.

## CHILD ROUTE

This release makes the six-question authority chain very clear. It also teaches an important lesson: a small flag may answer one narrow question, but it is not the whole state.

## What was added

- A finite grammar and typed state describe the query world.
- Local radial-angular support removes irrelevant candidates before expensive work.
- Compatibility, guard and confidence decide whether a candidate becomes a verified event.
- Packed records, direct Vulkan/SPIR-V compute and physical-device mappings are documented with precision rules.

## WHY IT MATTERS

Useful for query engines, packed event processing, hardware architecture studies and evidence-aware performance work.

## ADULT / VERIFICATION LANE

197 normalized mechanisms and 18 dependency-free tests are reported. Direct Vulkan execution was measured on SwiftShader, a CPU/software Vulkan device; the report explicitly marks `physical_gpu_claim=false`.

## HONEST LIMIT

It does not prove physical-GPU speed, zero heat, zero memory, universal  $O(1)$ , infinite detail or complete state in one bit.

**Source note:** Unified Geometric-Topological Substrate - GPU-Native Addendum, pp. 2-15 and Appendix D; created 15 Aug 2026.

# 2.0 FOUNDATION / PATTERN, KINEMATICS AND DYNAMICS

The toolbox gains shapes, paths, motion and bounded dynamics.

## CHILD ROUTE

Now the substrate can talk about many pattern families, fields, surfaces, routes, moving frames, timing and hybrid events while keeping the same six-question change rule.

## What was added

- Parametric patterns include superellipses, superformula families, rolling curves, spirals and splines.
- Field and surface families include superquadrics, tubes, helicoids, catenoids, periodic fields and arrival-time fields.
- Topology records winding, covering, route classes, braid words, cell incidence and chart consistency.
- Kinematics adds SE(2)/SE(3), moving frames, arc length, curvature limits, jerk-bounded schedules and articulated chains.
- Dynamics adds bounded integration, damped modes, diffusion and hybrid automata with degeneracy handling.

## WHY IT MATTERS

Useful for path planning, procedural shapes, simulation prototypes, motion design, calibration routes and rule-based geometry lessons.

## ADULT / VERIFICATION LANE

The release appends 60 engineering mechanisms, M198-M257, to the 197-mechanism baseline and reports 47 passing tests. Numerical and physical certification remains application-specific.

## HONEST LIMIT

It is still not a renderer. Mathematical names do not prove material behavior, self-assembly, energy properties or product readiness.

**Source note:** UGTS-KC 2.0 - Pattern, Kinematic Calculus and Dynamics Expansion, pp. 3-15; created 16 Aug 2026.

# 3.0 RUNTIME / TWO HANDS

The rule-world becomes an interactive scene with stable objects and replay.

## CHILD ROUTE

This release adds the parts needed to build and touch a graphics or game scene: assets, nodes, bounds, compiled geometry, materials, two-hand input, proposals, deterministic commit and replay.

## What was added

- Stable node identity is separated from asset identity, coordinates and lineage.
- Conventional AABB/BVH indexing works before semantic support and compatibility.
- Curves, strokes, swept tubes and bounded field meshes carry explicit approximation budgets.
- Left and right hand states, pinch hysteresis, bimanual transforms, ownership and handover are typed.
- CPU or GPU systems may propose events in parallel; verification and deterministic commit remain authoritative.

## WHY IT MATTERS

Useful for interactive graphics, editors, games, replayable manipulators, training tools and later physical or XR adapters.

## ADULT / VERIFICATION LANE

The release adds 72 mechanisms, M258-M329, and reports 117 passing tests: 47 retained from 2.0 plus 70 new. A runnable vertical slice exports SVG, glTF, USDA, scene and replay records.

## HONEST LIMIT

It is a tested reference vertical slice, not a finished commercial engine, full physics solver, standards-conformance suite or physical-GPU benchmark.

Source note: UGTS-KC Two Hands 3.0, pp. 2-16; created 16 Aug 2026.

## 3.6 FORMAL PROFILE / LITERAL REFERENTIAL

The rule cards move inside the world and can explain their dependencies.

### CHILD ROUTE

A definition is no longer only prose outside the system. Definitions receive stable IDs, types, dependencies, phases, provenance and content hashes. Instances point to those definitions, and the evaluator records the order used.

### What was added

- Literal definition nodes carry domain, codomain, parameters, dependencies, phase, invariants and SHA-256 content address.
- Unknown references, duplicate IDs, bad hashes and undeclared cycles are rejected.
- Hinges unify continuous rotation, discrete route selection and structural connector nodes.
- A ten-phase operation schedule makes parse, transform, support, compatibility, guard, transition and lineage order visible.
- A bounded Dutch 0-99 profile demonstrates that numeric place order and spoken order can be separate charts.

### WHY IT MATTERS

Useful for explainable rule systems, inspectable educational tools, content-addressed pipelines, authoring systems and traceable games.

### ADULT / VERIFICATION LANE

The package reports 36 v3.6 delta operators and 36 passing tests. It uses a separate K36 namespace rather than silently renumbering the 197-mechanism baseline.

### HONEST LIMIT

Language, binary and geometric feature-count coincidences remain metadata. They are not rewritten as numeric identity, hidden causal law or physical proof.

**Source note:** UGTS-KC 3.6 - Literal Referential Substrate, pp. 1-17; created 17 Aug 2026.

# 3.6.1 THE EMPTY SHELF CARD

Referenced intermediate artifact - title and contents unavailable

## WHAT WE KNOW

Later material refers to a 3.6.1 intermediate between the literal-referential 3.6 release and SCLP 3.6.2. The actual artifact is not among the current source files.

What a trustworthy library does

- Print the gap instead of closing it with a guess.
- Keep a placeholder catalog record with status: referenced, missing, unaudited.
- Do not assign a title, feature list, mechanism count or test count without the document.
- Update the family tree only when the source is recovered and checked.

## CHILD EXPLANATION

Sometimes a series says “Book 5 continues Book 4,” but Book 4 is missing from the shelf. We leave a card in its place so nobody thinks we forgot.

## ADULT CATALOG RECORD

Canonical placeholder: UGTS Formal Profile / Unresolved 3.6.1 Artifact @3.6.1. Relationship: referenced intermediate. Status: missing release placeholder.

## NO INVENTED CONTENT

This volume will not teach 3.6.1 as if it were known. It will only teach the historical fact that the gap exists.

### 3.6.2 DOMAIN PROFILE / SCLP

A small address can point to a finite map, but it cannot contain infinite detail.

#### CHILD ROUTE

SCLP combines a finite cone, paired sphere supports, a log-polar chart, bounded one-bit jitter, time phase and winding, explicit topological wraps, constraint release, a finite grammar and two exact 64-bit key layouts.

#### What was added

- A cone uses slant length plus half-angle; the exact primitive relation is separated from the certified sweep interval.
- Log-polar coordinates include a metric, Jacobian, velocity and acceleration, plus an explicit singular core.
- One-bit jitter receives an interval certificate and may not cross the event guard margin.
- Source half-turn and reflective Klein profiles are stored as different maps.
- A missing shackle becomes a constraint-row deletion with a measured nullity gain.
- Contiguous and Morton 64-bit keys are distinct, invertible layouts with finite capacity.

#### WHY IT MATTERS

Useful for compact indexing, geometric queries, finite lookup tables, bounded procedural structures and lessons about quantization and uncertainty.

#### ADULT / VERIFICATION LANE

SCLP reports 50 atomic delta operators, 22 content-addressed definitions, 70 new tests and 153 retained-plus-new passing tests.

#### HONEST LIMIT

A 64-bit key is an index, not an entire world. Width ratios are not equal-information compression without reconstruction and error equivalence. Generic sweeps are not automatically exact SDFs.

Source note: UGTS-KC 3.6.2 SCLP, pp. 1-15; created 17 Aug 2026.

### 3.6.3 DOMAIN PROFILE / SARA

Public information is not permission, and a checksum is not ownership.

#### CHILD ROUTE

SARA maps public deterministic-wallet standards into typed definitions and public-only certificates. For younger readers, the core lesson is simple: seeing a public address does not give permission to find, use or transfer its secret key.

#### What was added

- BIP39 entropy, checksum and 11-bit word cells form an ordered word path.
- BIP32 normal and hardened derivation edges remain different types.
- BIP84, public keys, HASH160, witness programs, scripts, Bech32 and Bech32m are represented as public relations.
- An authorization guard admits public vectors, self-owned known inputs, watch-only checks and test networks.
- Secret fields are transient and prohibited from serialized certificates and lineage.
- Search-space estimates are calculated without candidate generation.

#### WHY IT MATTERS

Useful for standards conformance, public address decoding, watch-only consistency, custody-process checks, test-network education and permission/safety lessons for older learners.

#### ADULT / VERIFICATION LANE

SARA reports 45 new operators, 19 content-addressed definitions, 42 new tests and 195 combined passing tests. The reference runtime has no network or transaction capability.

#### HONEST LIMIT

No mnemonic enumeration, passphrase spraying, private-key scan, target-address search, signing, broadcast, balance targeting or funds-transfer path is included or taught.

**Source note:** UGTS-KC 3.6.3 SARA, pp. 1-15; created 18 Aug 2026. Place on an older-learner/adult-guided shelf.

# 3.9 APPLICATION / ELIZABETH VECTOR GAME

The ideas become a small playable vector world.

**CHILD ROUTE**

KC Elizabeth adds a complete 2D game-making path: editable project records, vector art, named actions, animation, tilemaps, procedural sound, a deterministic fixed-step game world, collisions, saves and a self-contained HTML5 build.

## What was added

- Vector assets use paths, paints, gradients, primitive factories, SVG inspection and direct Canvas rendering.
- Keyboard, pointer, gamepad and touch inputs resolve into named deterministic action frames.
- Animation supports tracks, easing, clip modes, crossfades and state machines.
- Tilemaps support ASCII import, reachability, A\* paths and merged colliders.
- GameWorld uses explicit phases for input, controller, physics, bounds, collision, gameplay, update, camera and flush.
- The offline browser target provides Canvas, Web Audio, controls, persistence and diagnostics without external runtime libraries or network assets.

|                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WHY IT MATTERS</b> <p>Useful for games, interactive lessons, vector animation, classroom projects, accessible input experiments and later physical-game companions.</p>                           | <b>ADULT / VERIFICATION LANE</b> <p>The release adds 60 mechanisms, M330-M389, bringing its combined catalog to 389 and its retained-plus-new test suite to 225. Elizabeth's Vector Garden is an editable project with a deterministic headless smoke run and offline browser build.</p> |
| <b>HONEST LIMIT</b> <p>No native 3D renderer, general rigid-body/joint/deformable engine, skeletal suite, multiplayer service, console SDK, OpenXR binding or security certification is claimed.</p> |                                                                                                                                                                                                                                                                                          |

Source note: UGTS-KC 3.9 - KC Elizabeth Vector Game Runtime, pp. 2-16; created 23 Aug 2026.

# WHAT STAYS THE SAME ACROSS THE WHOLE FAMILY

Six permanent promises

|                                                                                                                                                        |                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>LOCAL FIRST</b><br>Do not ask the whole world when a smaller support can reject irrelevant work.                                                    | <b>MATCH BEFORE CHANGE</b><br>Same place is not enough; types, sheets, phases, roles and policies matter.      |
| <b>BOUNDARIES ARE TYPED</b><br>A zero, threshold, time or signal must have an explicit meaning and error rule.                                         | <b>PROPOSE, THEN COMMIT</b><br>Detection may be parallel, but authoritative change is verified and ordered.    |
| <b>IDENTITY NEEDS HISTORY</b><br>Coordinates, hashes and compact keys do not replace lineage.                                                          | <b>EVIDENCE HAS LIMITS</b><br>Tests, demos and measurements establish only what their declared setup supports. |
| <b>CHILD SUMMARY</b><br>Find the right place. Check the match. Wait for the real moment. Check the whole idea. Change one clear thing. Keep the story. |                                                                                                                |

# WHAT GROWS OVER TIME

The backbone stays; the vocabulary and applications expand

## WHAT THE LIBRARY ADDS, STEP BY STEP

|       |                         |                                                                    |
|-------|-------------------------|--------------------------------------------------------------------|
| 1.1   | <b>CHECK</b>            | Where, match, ready, checked, change, remember.                    |
| 2.0   | <b>DESCRIBE</b>         | Patterns, fields, routes, motion and bounded dynamics.             |
| 3.0   | <b>INTERACT</b>         | Scenes, assets, two hands, proposals, replay and exports.          |
| 3.6   | <b>EXPLAIN ITSELF</b>   | Definitions, dependencies, hashes, hinges and operation order.     |
| 3.6.2 | <b>PACK &amp; INDEX</b> | Finite geometry, log-polar charts, guard intervals and exact keys. |
| 3.6.3 | <b>ASK PERMISSION</b>   | Public certificates, authorization and no-secret boundaries.       |
| 3.9   | <b>PLAY</b>             | Vector art, input, collision, audio, saves and offline HTML5.      |

### DO NOT ADD BRANCH COUNTS BLINDLY

Mechanism and test counts belong to particular packages and retained suites. The 3.6 branch uses its own delta namespaces; SCLP and SARA retain different test baselines; 3.9 continues the M001-M389 application catalog. Counts are traceability tools, not one universal score.

# WHERE CAN THESE IDEAS BE USED?

Applications stay tied to the versions that actually support them

## PLAYABLE LEARNING

Six-question card games, Definition Breaker, vector games, cooperative rule puzzles.

Source spine: 1.1 + 2.0 + 3.0 + 3.6 + 3.9

## INTERACTIVE GRAPHICS

Stable scenes, vector art, asset instances, two-hand transforms, offline browser output.

Source spine: 3.0 + 3.9

## MOTION & CONTROL STUDIES

Paths, moving frames, timing, bounded dynamics, guard events and articulated chains.

Source spine: 2.0

## EXPLAINABLE RULE SYSTEMS

Definitions, dependencies, hashes, traces, operation order and replay.

Source spine: 3.6 + 3.0

## PACKED MAPS & QUERIES

Finite keys, quantization, log-polar charts, guard intervals and bounded grammars.

Source spine: 1.1 + 3.6.2

## PUBLIC DATA & PERMISSION

Public format validation, watch-only checks, authorization and secret-egress rules.

Source spine: 3.6.3 - older learner/adult guided

## APPLICATION RULE

A source may inspire an application only where its documented strengths fit. Unrelated domain profiles should not be imported merely because they share referential language.

# THE COLLECTION PLAN

Separate shelves prevent the old mixed-sheet problem from returning

## THE FUTURE LIBRARY HAS FOUR SHELVES

History, concepts, applications and adult reference stay connected but do not get mixed together.

### A. CHRONOLOGICAL HISTORY

One slim volume per release, in date and lineage order.

A0 Map | A1 Compute | A2 Foundation | A3 Runtime | A4 Formal | A5 SCLP | A6 SARA | A7 Elizabeth

### B. CONCEPT LAB

Six big questions, shapes, motion, topology, definitions, evidence and replay.

B1 Where? | B2 Match? | B3 Ready? | B4 Checked? | B5 Change | B6 Remember

### C. APPLICATION STUDIO

Games, physical kits, interactive scenes, motion, indexing and public-data safety.

C1 Definition Breaker | C2 Vector Garden | C3 Two Hands Lab | C4 Packed Map Lab | C5 Public & Permission

### D. ADULT REFERENCE

Formal notation, schemas, catalogs, source audits, validation reports and implementation notes.

D1 Glossary | D2 Version ledger | D3 Operator catalog | D4 Evidence handbook | D5 Educator guide

## VERSIONING RULE FOR THE LIBRARY

Library volumes receive their own clear edition number, separate from the substrate release they explain. Example: UGTS Library A3, edition 0.1, may explain UGTS Runtime / Two Hands 3.0.0 without pretending to be substrate version 3.0.1.

# FIRST LIBRARY ACTIVITY: BORROW THE SHAPE

A cooperative tabletop lesson for one or many players

## Materials

- Six question cards from the next pages.
- Three safe objects with different shapes or textures.
- A tray or marked “WHERE” area.
- Two matching labels, symbols or tactile keys.
- Six beads, blocks or marks for the memory trail.

## Play

5. Place one object in the WHERE area and keep the others outside.
6. Choose a MATCH rule, such as “only circles” or “only the card with the same texture.”
7. Choose a READY rule, such as a timer, a bell, a raised ridge or a partner signal.
8. A player proposes moving the object. The group uses CHECKED? to review WHERE, MATCH and READY.
9. On CHANGE, move exactly one object or update exactly one label.
10. On REMEMBER, add one bead or mark that records what happened and why.

### ACCESS OPTIONS

A player may answer by eye gaze, point, sign, AAC, switch, sound, matching object, partner-assisted scanning or written choice. No speech is required.

### EXTENSION FOR OLDER LEARNERS

Add two simultaneous proposals. Give them priorities. Decide whether they commute, conflict or require rejection. Record the reason and replay the sequence.

### CONSENT RULE

Touching another person or their mobility, communication or sensory equipment is never a game requirement. Use objects, agreed hand-over-hand support or partner-assisted scanning only with consent.

# PRINTABLE QUESTION CARDS - SET 1

Cut on the outer border. Color is repeated by number, shape and text.

|                                                                                                                                                                                                  |                                                                                                                                                                       |                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1</b></p> <p><b>WHERE?</b></p> <p><b>SUPPORT</b></p> <p>Where can this rule matter?</p> <p><b>TACTILE IDEA</b><br/><b>DOT FIELD</b></p>                                                    | <p><b>2</b></p> <p><b>MATCH?</b></p> <p><b>COMPATIBILITY</b></p> <p>Are these pieces allowed to work together?</p> <p><b>TACTILE IDEA</b><br/><b>PAIRED NOTCH</b></p> | <p><b>3</b></p> <p><b>READY?</b></p> <p><b>GUARD</b></p> <p>Has the needed boundary, time or signal arrived?</p> <p><b>TACTILE IDEA</b><br/><b>SINGLE RIDGE</b></p> |
| <p><b>PROTOTYPE NOTE</b></p> <p>Final tactile symbols require co-design and validation for spacing, material, braille, contrast, durability and safe use. These labels are scaffolding only.</p> |                                                                                                                                                                       |                                                                                                                                                                     |

# PRINTABLE QUESTION CARDS - SET 2

Keep the sequence number with every card

|                                                                                                                                                                                        |                                                                                                                                  |                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>4</b></p> <p><b>CHECKED?</b></p> <p>VERIFIED EVENT</p> <p>Did the whole proposal pass its checks?</p> <p>TACTILE IDEA<br/>STAR / BURST</p>                                       | <p><b>5</b></p> <p><b>CHANGE</b></p> <p>TRANSITION</p> <p>What exact thing changes now?</p> <p>TACTILE IDEA<br/>ARROW GROOVE</p> | <p><b>6</b></p> <p><b>REMEMBER</b></p> <p>LINEAGE</p> <p>What must be kept so we can replay it?</p> <p>TACTILE IDEA<br/>LINKED RINGS</p> |
| <p><b>MEMORY TRAIL</b></p> <p>Use beads, linked rings, stamped shapes, recorded words or a digital event list. The form may change; the order and meaning must remain recoverable.</p> |                                                                                                                                  |                                                                                                                                          |

# CORE GLOSSARY

Child language and formal language refer to the same architecture

| CHILD TERM      | FORMAL TERM                | MEANING                                                                                 |
|-----------------|----------------------------|-----------------------------------------------------------------------------------------|
| WHERE?          | Support                    | Whether a rule or object is locally relevant to the query.                              |
| MATCH?          | Compatibility              | Whether types, sheets, phases, roles and policies may interact.                         |
| READY?          | Guard                      | Whether a boundary, timing, threshold or confirmation condition is accepted.            |
| CHECKED?        | Verified event             | A complete proposal that passed the declared checks and conflict policy.                |
| CHANGE          | Transition                 | An atomic state patch applied after verification.                                       |
| REMEMBER        | Lineage                    | Ordered event history, identity, checkpoints and replay.                                |
| RULE CARD       | Definition node            | A typed, versioned, content-addressed rule record.                                      |
| NEEDS           | Dependency                 | A definition or object required before another can be evaluated.                        |
| TRY WORLD       | Proposal / candidate state | A reversible suggestion kept separate from authoritative state.                         |
| SHEET / ROUTE   | Topology state             | A gluing, layer, winding or route class that may distinguish co-located states.         |
| PICTURE         | Projection                 | A render, export or display that helps us see but is not automatically state authority. |
| TEST / EVIDENCE | Validation result          | A bounded statement about what a declared setup checked.                                |

# SOURCE AND EVIDENCE REGISTER

Primary technical shelf used for this first installment

| SOURCE                   | DATE        | PP. | USED FOR                                                                          | SHA-256 PREFIX   |
|--------------------------|-------------|-----|-----------------------------------------------------------------------------------|------------------|
| UGTS-GN 1.1              | 15 Aug 2026 | 28  | Canonical authority chain; packed precision; performance and evidence boundaries  | 3a6b6c57118ca4e2 |
| UGTS-KC 2.0              | 16 Aug 2026 | 23  | Patterns, fields, topology, kinematics, dynamics and event degeneracy             | a29dd9959ebd300c |
| KC Two Hands 3.0         | 16 Aug 2026 | 20  | Stable scene identity, two-hand interaction, proposal/commit, replay and export   | 183ffa4e2bf29cfa |
| UGTS-KC 3.6              | 17 Aug 2026 | 19  | Content-addressed definitions, dependencies, hinges and explicit order            | e21433cd8378368d |
| UGTS-KC 3.6.2 SCLP       | 17 Aug 2026 | 19  | Finite geometry, log-polar calculus, intervals, topology, grammar and 64-bit keys | 6987490a0a7b7ed7 |
| UGTS-KC 3.6.3 SARA       | 18 Aug 2026 | 18  | Public deterministic-wallet standards, authorization and secret-egress boundaries | c5a8256e345047d5 |
| UGTS-KC 3.9 KC Elizabeth | 23 Aug 2026 | 16  | Vector art, input, game world, collisions, audio, saves and offline HTML5         | d7ee3c91abf60e35 |

## SOURCE GAPS

The original UGTS Core / Baseline 1.0 source is referenced but not present in this shelf. UGTS 3.6.1 is referenced but missing. Both remain explicit catalog gaps.

## PRIVACY AND ATTRIBUTION

The technical reports contain requester-supplied attribution and identifiers. This child-facing library edition omits unnecessary personal identifiers. Source attribution and legal boundaries remain available in the adult reference shelf.

# INSTALLMENT 1 COMPLETE

The map is stable enough to begin the books

What this installment establishes

- A child-to-expert language bridge centered on the six big questions.
- A chronological ledger that separates dates from actual package lineage.
- A visible split between the application spine and formal/domain branch.
- A protected placeholder for the missing 3.6.1 artifact.
- A future collection structure that separates history, concepts, applications and adult reference.
- A first cooperative activity and printable prototype cards.

The next volume in the work sequence

## VOLUME 1 - THE SIX BIG QUESTIONS

A full child-first book with stories, tabletop rooms, tactile/AAC scaffolds, worked examples, incorrect-order puzzles, educator notes and direct bridges to support, compatibility, guard, verified event, transition and lineage.

Open audit items

- Recover and audit the referenced UGTS Core / Baseline 1.0 document if it becomes available.
- Recover or formally close the 3.6.1 gap.
- Co-design child, tactile, braille, AAC and easy-read versions with actual users and librarians.
- Separate source-derived statements from future educational design in every later volume.

## EDITION CONTROL

UGTS Library Edition, Volume 0, edition 0.1. This is an educational organization layer. It does not modify the formal substrate versions it explains.